Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/10/2006
PI Name:	Gatley, Samuel J Ph.D.		
Project Title:	MicroPET Studies of Brain Damage by Heavy Ion Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Zip Code:	02115	**Congressional District:**	8
Comments:	Formerly at Brookhaven National Laboratory; moved sometime in 2005.		
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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Performance Goal No.:			
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Task Description:	Space travel beyond the Earth's protective magnetic field (for example, to Mars) will involve exposure of astronauts to irradiation by high energy nuclei such as 56Fe (HZE radiation) which are a component of galactic cosmic rays. These particles have high linear energy transfer (LET) and are expected to irreversibly damage cells they traverse. Exposure to HZE radiation may therefore cause progressive deterioration of brain function, adding to damage involved in normal aging. We propose a study of the hypothesis that long-term alterations in brain function are induced after exposure of the brain to 1 GeV/n iron-56 particles at doses in the region of 100 cGy. Previous published studies support this notion but are not definitive, especially with regard to long-term effects. We will measure: (a) Cocaine induced locomotor activity in rats at times up to two years after irradiation. We have shown in mice that the test is reproducible and repeatable so that the temporal evolution of the altered behavior can be followed. (b) Tissue levels of dopamine in striatum, since this is a sensitive measure of nigrostriatal degeneration in humans. (c) The ability of L-deprenyl, a monoamine oxidase-B inhibitor with neuroprotective properties, to mitigate effects of HZE radiation seen in (a) and (b).		

	We will also conduct longitudinal microPET studies of the rat brain over the same timescale of the behavioral studies. These will set the groundwork for eventual PET studies of non human primates which, with their more complex brains, may be more relevant to the situation of astronauts. PET scanning after missions could provide early detection of subtle brain damage. Scanning could also permit screening for individuals likely to be at particular risk on prolonged missions. To determine whether heavy ion irradiation of the mammalian brain causes persistent damage in behavioral and neuroimaging studies. These studies will help determine the risks of neurological damage from the space radiation environment during deep space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Introduction: The objective of the research is to use microPET imaging as well as behavioral testing to evaluate the effects of irradiation with iron-56 particles on the rat brain. The studies will help address Critical Questions 10.05 (Are there unique biological effects associate with HZE's?) and 10.07 (How can animal experiments…extrapolate the human risk…?), of the Critical Path Roadmap. The background to this grant proposal is that in studies in mice funded by the NSBRI, the PI and Dr Vazquez had shown that cocaine stimulated locomotor activity was reduced by a moderate dose (120 cGy) of 1 GeV 56Fe particles. We postulated that imaging experiments in animals might provide more sensitive and earlier indicators of damage due to HZE particles than behavioral tests. Since the small size of the mouse brain is not well suited to the spatial resolution offered by microPET, we proposed to repeat some of our studies in a rat model. We anticipated that this would enable us to identify imaging correlates of behavioral endpoints. A specific hypothesis was that changes in the metabolic rate for glucose in striatum of animals tested several months after irradiation would be correlated with alterations in locomotor activity. Other behavioral tests would be conducted as time and opportunity allowed while animals were between imaging sessions. Our experimental plan also included evaluation of whether the neuroprotective drug L-deprenyl would reduce the effect of radiation on locomotor activity. In addition, we planned to conduct microPET studies of brain monoamine oxidase A and monoamine oxidase B in rats before and at various times after irradiation with HZE particles. The radiotracers to be used are [11C]clorgyline for MAO-A and L-[11C]deprenyl-d2 for MAO-B; the PET group at Brookhaven has extensive experience with these radioactive drugs. The hypothesis is that MAO-A, which is located in nerve terminals, will be unchanged or decreased after irradiation, while MAO-B, which is located in glial cells, will be increased after irradiation. Neurochemical effects that could be measured using PET could in principle be applied in astronauts, in terms of detecting and monitoring subtle neurological damage that might have occurred during long space missions. Irradiations: Animals (60 per run) have received doses of 600 MeV/n iron-56 collimated to the head region during NSRL-2 and NSRL-4. Another batch of 60 will be similarly irradiated during NSRL-5. Imaging: Methods– PET Studies. Animals were irradiated at NSRL during April 2004 (ie NSRL-2). The images shown are from MicroPET experiments in which rats were given an i.p. injection of FDG (approximately 700 uCi), placed in a novel environment for 45 minutes during the radiotracer uptake period, anesthetized and then scanned for 20 minutes. The following day, animals were irradiated at 0, 120, or 240 cGy with HZE Fe56 at the NSRL. One week, three months and eleven months later, the animals were scanned again. The images were co-registered with a rat brain atlas to define brain regions. Results - There were no significant decreases in either absolute values (nCi/cc) or regional ratios for the NSRL-2 rats. (See figures and images). Since we are interested primarily in long-term effects of HZE radiation, studies with the NSRL-4 rats are still pending. Discussion - Animals irradiated at 120 cGy and 240 cGy have not yet demonstrated any significant decreases in glucose metabolism as measured by MicroPET, though it is possible that group differences will become apparent at longer periods after irradiation. Another possibility is that there are differences, but that they only occur in brain regions smaller than the resolution of microPET. In fact, some recent papers have indicated that drug induced alterations in locomotor activity are associated with significant changes in glucose metabolism only in the nucleus accumbens, a small region of the striatum. If this is true for HZE radiation, we will not be able to detect the decreases with the microPET. Therefore, after completing behavioural and PET studies we will conduct quantitative ex vivo autoradiographic studies of local rates of glucose metabolism using carbon-14 labeled 2-deoxyglucose which is able to resolve very small brain areas including the accumbens.
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